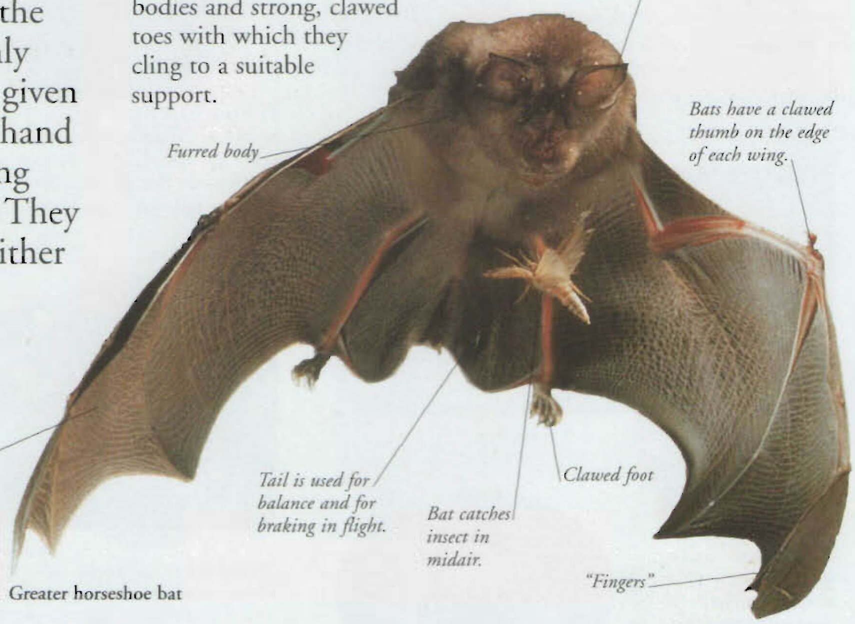
BATS

WITH ALMOST 1,000 species, bats are the second largest order of mammals after the rodents. They are the only mammals that can truly fly. The name given to their order is Chiroptera, meaning "hand wings". When bats are resting, they hang upside-down. Most bats are nocturnal. They eat a variety of food, which they find either by scent and sight, as fruit bats do, or by using sound waves, a process called echolocation, as insect-eating bats do.

Bat features

A bat's wing consists of an elastic membrane of skin that is stretched between the elongated fingers of its front limb, and back to its hind limb. Bats have lightweight bodies and strong, clawed toes with which they cling to a suitable support.



Greater horseshoe bat

Types of bat

Bats are divided into two groups. These are the Megachiroptera, or megabats, which are the old world fruit bats, and the Microchiroptera, or microbats, sometimes called insect-eating bats.

Megabats

Fruit bats, or megabats, are also sometimes called flying foxes. They live in the tropical and subtropical parts of Africa, Asia, and Australasia. Most megabats eat fruit, but some also feed on flowers, nectar, and pollen.

Epauletted fruit bat

Long-eared bat

Microbats

The term insect-eating bats is a misleading name for these bats. Many feed on fruit, meat, fish, pollen, and even blood, as well as insects. Microbats live in both temperate and tropical regions, but in cooler climates they hibernate or migrate for the winter.

Roosts

Bats need a variety of places to roost, or rest. At night they rest between bouts of feeding and often settle to eat large prey. During the day, they need somewhere to sleep and groom. Females choose a safe, warm place to give birth.

Cave habitats

In warm climates, caves provide daytime and nursery roosts, where females give birth and look after their young. Bracken Cave in Texas, USA, has the largest colony in the world with up to 20 million bats.

Free-tailed bats in Bracken Cave

Tent bats

Tree habitats

Microbats often roost in tree holes, such as old woodpecker nests, or cracks caused by storm damage. These Honduran white bats, also called tent bats, build a tent from large leaves.

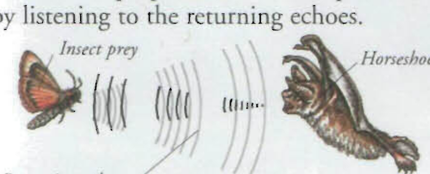
Hibernation

Bats need to hibernate somewhere cold but where they will be protected from frost, which would kill them. The place where they roost, called a hibernaculum, also has to be damp so that the bats do not dry out. Suitable sites include caves, loft spaces, and tree holes.

Natterer's bat

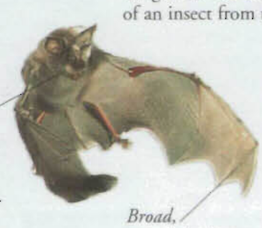
Echolocation

To find objects in the dark, a microbat makes bursts of high-frequency sound. The sound bounces off objects, such as a moth, and the bat pinpoints the moth's position by listening to the returning echoes.



How a horseshoe bat catches prey

1 The "horseshoe" on the bat's nose focuses the sound into a narrow beam. The bat sweeps its head from side to side as it flies along, scanning for insects.



2 The bat's large, mobile ears pick up vibrations made by the movement of an insect's wings. The bat can tell the size of an insect from the vibrations.

3 When the bat has located its prey, it scoops up the insect in its wings, often eating in midair.